



Development of Model Predictive Controlled 12-DoF Bipedal Robot

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ABSTRACT

State-of-the-art push recovery algorithms for bipedal robots frequently demand prohibitively expensive and computationally intensive hardware. To address these constraints, this study introduces a two-phase control architecture tailored for low-budget bipedal systems. Grounded in the Linear Inverted Pendulum Model (LIPM), the framework generates nominal walking trajectories offline by employing Zero-Moment Point (ZMP) preview control, a derivative of Model Predictive Control (MPC). For the preservation of balance during walking, real-time push recovery is facilitated by an online Linear Quadratic Regulator (LQR). This approach validates that stability against external perturbations can be effectively achieved utilizing resource-constrained, low-cost microcontrollers.

Keywords: Bipedal Robotics, Push Recovery, ZMP Preview Control, Low-Budget Robotic Platforms.



1. INTRODUCTION

The control of bipedal robots remains one of the most challenging endeavors in robotics due to their high degrees of freedom and inherently unstable dynamics. To maintain stable walking, researchers have traditionally utilized the Zero-Moment Point (ZMP) criterion [Vukobratovic, 2004] and the Linear Inverted Pendulum Model (LIPM) to simplify complex multibody dynamics into a manageable linear model [Kajita, 2014]. While these fundamental methods provide a robust foundation for nominal gait generation, maintaining stability under external disturbances—commonly referred to as push recovery—requires advanced control strategies capable of reacting to instantaneous deviations.

In recent years, Model Predictive Control (MPC) [Wieber, 2006] has been widely utilized as a successful strategy for dynamic walking and push recovery by optimizing future states over a receding time horizon. However, the continuous real-time execution of MPC requires a massive computational load, necessitating heavy and expensive onboard computers. Such demands impose a severe limitation on low-budget bipedal platforms designed around edge computing devices. Because entry-level microcontrollers, such as the Teensy 4.1, are incapable of solving complex optimization matrices in real time, an alternative approach is required. To overcome these hardware limitations, this paper utilizes a ZMP-based preview control [Kiyotsugu Takaba, 2003] formulation to generate walking trajectories offline. These trajectories are subsequently stored and deployed directly onto the microcontroller.

On the hardware level, an online Linear Quadratic Regulator (LQR) is implemented to maintain dynamic balance while the robot executes the pre-computed walking trajectory. When an external impact occurs, resulting in a physical deviation from the reference path, an Inertial Measurement Unit (IMU) is utilized to continuously record the robot's real-time body acceleration and angular velocity to precisely quantify the disturbance. Instantaneous deviations in the Center of Mass (CoM) position and velocity are then computed based on these sensor readings. To minimize this error and restabilize the system, an offline-computed optimal feedback gain matrix derived from the LQR formulation is multiplied by the state error. This computationally efficient operation allows the microcontroller to swiftly generate the corrective control inputs necessary for robust push recovery.



2. MODELLING

Figure 1 illustrates the hardware topology and kinematic chain of the 12-DoF bipedal platform. The block diagram on the left details the parent-child hierarchy from the floating base (torso) to the end-effectors (feet), while the schematic layout on the right shows the physical placement and color-coded clustering of the joints. This bipedal robot is designed with a total of 12 degrees of freedom (DoFs), allocating 6 DoFs per leg to ensure full spatial mobility and stepping ability. The kinematic chain from the hip to the foot is defined in the following sequence:

Hip Yaw $\rightarrow$ Hip Roll $\rightarrow$ Hip Pitch $\rightarrow$ Knee Pitch $\rightarrow$ Ankle Pitch $\rightarrow$ Ankle Roll.

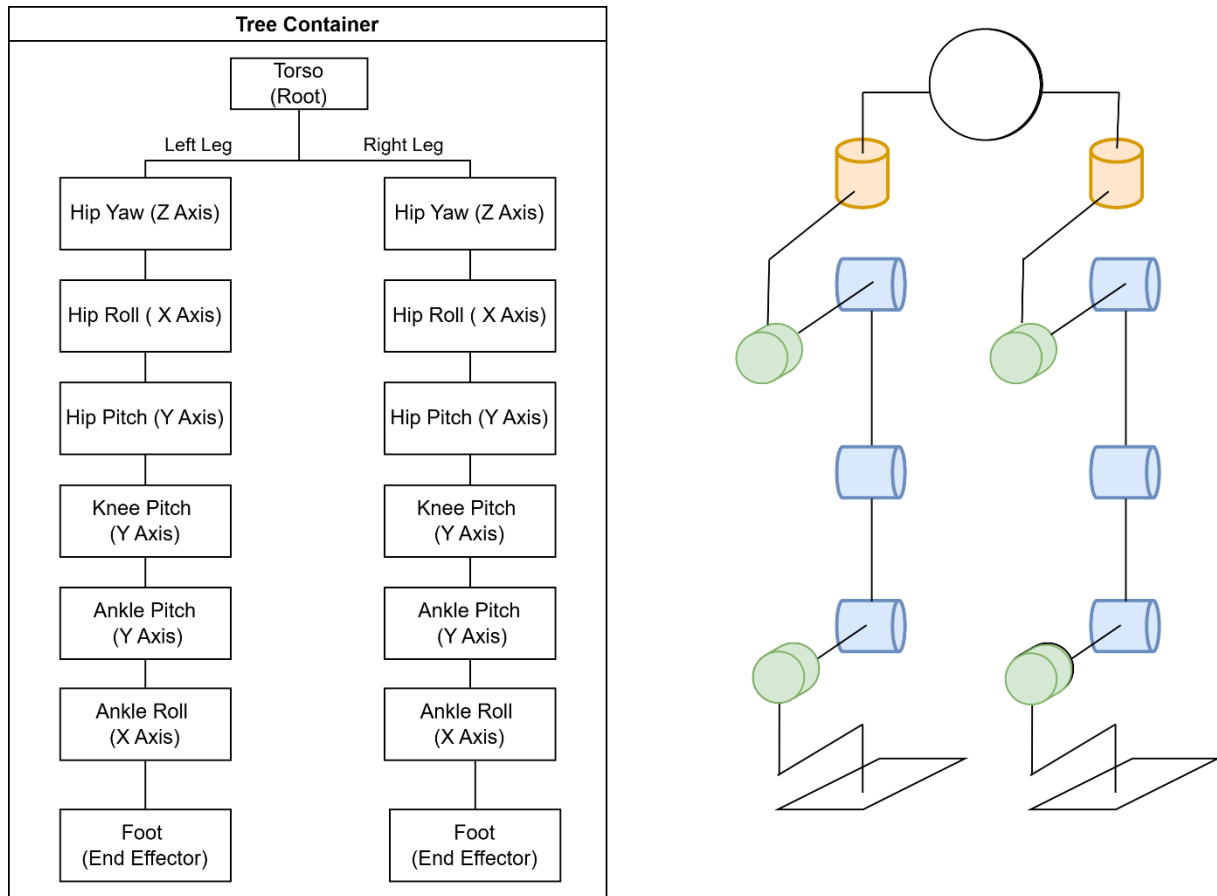


Figure 1: Kinematic Tree



The physical prototype incorporates an offset kinematics design. Stacking actuators directly upon each other may lead physical collisions and limits the range of motion. To avoid this, consecutive motors are structurally separated by using custom designed brackets. This configuration could convert decoupling kinematic since motor rotational axes crossing one imaginary point [Sun, 2011]. This mechanical reality is explicitly mapped into the Denavit-Hartenberg (D-H) parameters and the MuJoCo simulation environment to ensure minimize “Sim to Real” gap during inverse kinematics (IK) calculations [Kawaharazuka,2025]. Moreover specific Z offset is preserved between the ankle roll actuator and the flat foot to modelling the natural human ankle height, ensuring realistic torque distribution on the ground contact points.

Following the low cost architecture aim of this study, ST 3025 serial bus servo motors are placed across all joints. In bipedal locomotion, specially during Y-axis push recovery and squatting motions, the highest torque loads occur in the sagittal plane. Consequently, the structural integrity around the Hip Pitch and Knee Pitch joints is maximized. The physical links connecting the actuators are manufactured using CNC machined aluminum to optimize the strength to weight ratio. The leg geometry is symmetrically designed with a thigh length of 0.20 m and a shank length of 0.20 m. During nominal walking, the robot is constrained to keep its Center of Mass (CoM) height constant at 0.32 m. As a result, the low-level microcontroller can rapidly execute Capture Point step reflexes without exceeding the torque limits of the entry-level actuators.

3. METHODOLOGY

3.1 System Overview

The proposed control architecture is structured into two hierarchical layers to satisfy the real-time constraints imposed by low-cost embedded hardware. The upper layer performs all computationally intensive tasks offline: given a desired footstep sequence, the LIPM-based preview controller generates a Center of Mass (CoM) trajectory that satisfies ZMP stability constraints over a finite preview horizon. The corresponding joint-space trajectories are then computed via analytical Inverse Kinematics (IK) and stored as tables in the microcontroller’s memory to deployment.



The lower layer operates entirely online during locomotion. An MPU6050 Inertial Measurement Unit (IMU) continuously monitors the robot's body orientation and angular velocity. A complementary filter fuses these measurements to estimate the instantaneous CoM state deviation from the nominal trajectory. A pre-computed LQR feedback gain matrix then maps this state error to corrective joint commands, which are superimposed onto the nominal trajectory commands. Since the online computation reduces to a single matrix-vector multiplication, the architecture is fully compatible with the microcontroller operating at 600 MHz. Proposed system presented as overall block diagram [Fig. 2].

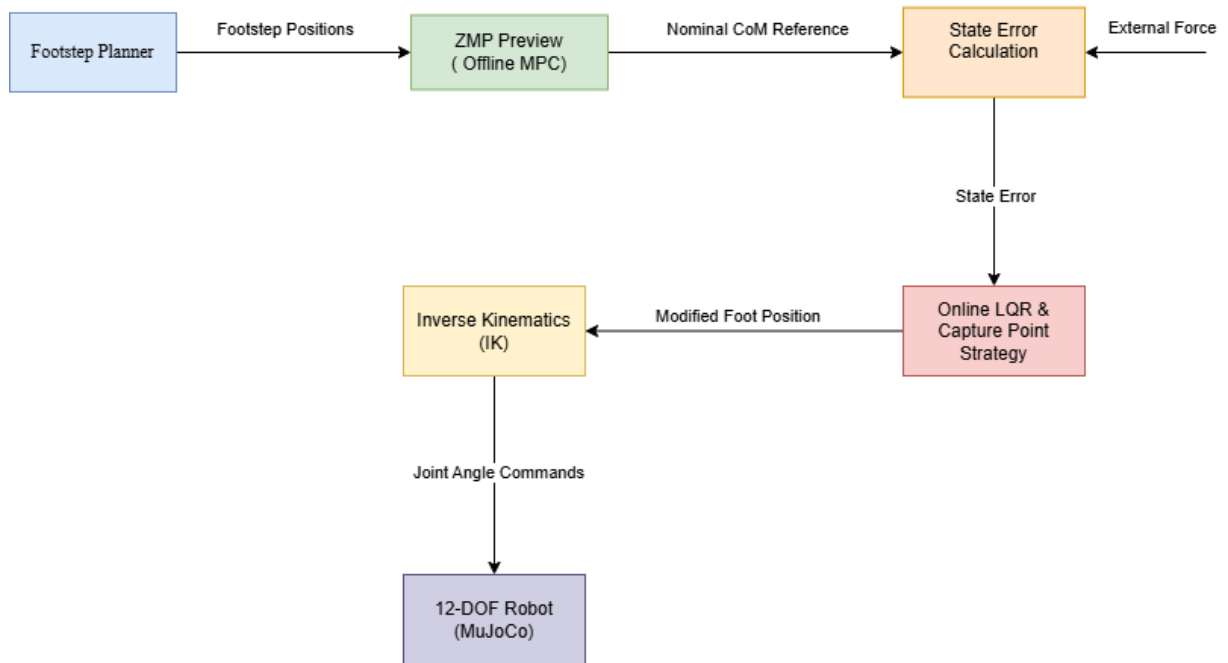


Figure 2: Overall Control Block Diagram

3.2 Linear Inverted Pendulum Model

The bipedal robot is modeled as a Linear Inverted Pendulum (LIP), in which the total robot mass m is lumped at a single CoM point constrained to move at a constant height (z_c) above the ground plane. Neglecting the angular momentum about the CoM, the equation of motion in the sagittal plane is derived from Newton's second law applied at the ZMP:

$$\ddot{x} = \frac{g}{z_c} (x - p_x) \quad (1)$$



where x is the horizontal CoM position, p_x is the ZMP position, and $g = 9.81 \text{ m/s}^2$ is gravitational acceleration. An identical formulation holds for the frontal plane with CoM position y and ZMP position p_y . The ZMP is defined as the point on the ground plane where the net ground reaction moment about the horizontal axes is zero. For a statically stable configuration, the ZMP must remain within the support polygon at all times [Kajita, 2003].

The key advantage of the LIPM is simplifying complex nonlinear equations into manageable linear forms. Linear equations require less computation load on application of optimal control theory. The constant height constraint z_c is maintained by coordinated ankle and knee joint actuation throughout the gait cycle.

3.3 Discrete-Time State-Space Formulation

For digital implementation, equation (1) is discretized with a sampling period T using a zero-order hold. Defining the state vector as:

$$\mathbf{x}_k = \begin{bmatrix} x_k \\ \dot{x}_k \\ \ddot{x}_k \end{bmatrix} \quad (2)$$

and selecting the CoM jerk $u_k = \ddot{x}_k$ as the control input, the discrete-time state-space representation becomes:

$$\mathbf{x}_{k+1} = \mathbf{A}\mathbf{x}_k + \mathbf{B}u_k \quad (3)$$

$$p_k = \mathbf{C}\mathbf{x}_k \quad (4)$$

where the system matrices are:

$$\mathbf{A} = \begin{bmatrix} 1 & T & \frac{T^2}{2} \\ 0 & 1 & T \\ 0 & 0 & 1 \end{bmatrix}, \mathbf{B} = \begin{bmatrix} \frac{T^3}{6} \\ \frac{T^2}{2} \\ T \end{bmatrix}, \mathbf{C} = \begin{bmatrix} 1 & 0 & -\frac{z_c}{g} \end{bmatrix} \quad (5)$$

The output represents the predicted ZMP position, which must track the desired ZMP reference throughout the motion.



3.4 Offline Trajectory Generation via ZMP Preview Control

The nominal CoM trajectory is generated using the preview control method of Kajita et al. [Kajita, 2003]. The method is formulated as a receding-horizon optimal control problem that minimizes the following quadratic cost function over a preview window of N steps:

$$J = \sum_{k=0}^{\infty} [Q_e \tilde{p}_k^2 + \mathbf{x}_k^T \mathbf{Q}_x \mathbf{x}_k + R u_k^2] \quad (6)$$

where $\tilde{p}_k = p_k - p_{ref,k}$ is the ZMP tracking error Q_e is the ZMP error weight Q_x is the state weight matrix and R is the control effort weight [Kajita, 2003].

To incorporate preview information, an augmented error system is constructed by introducing the integrator state:

$$e_{k+1} = e_k + \mathbf{C} \mathbf{A} \mathbf{x}_k + \mathbf{C} \mathbf{B} u_k - \Delta p_{ref,k+1} \quad (7)$$

The augmented state vector $\tilde{\mathbf{x}}_k = [e_k, \Delta \mathbf{x}_k^T]^T$ leads to the augmented system:

$$\tilde{\mathbf{x}}_{k+1} = \tilde{\mathbf{A}} \tilde{\mathbf{x}}_k + \tilde{\mathbf{B}} \Delta u_k - \tilde{\mathbf{G}} \Delta p_{ref,k+1} \quad (8)$$

The optimal control law that minimizes equation (6) is then:

$$\Delta u_k = -K_e e_k - \mathbf{K}_x \Delta \mathbf{x}_k + \sum_{j=0}^{N-1} f_j \Delta p_{ref,k+1+j} \quad (9)$$

where K_e and K_x are the optimal feedback gains obtained by solving the discrete-time Algebraic Riccati Equation (DARE) [Kayatama, 1995], and f_j are the preview gains that weight future ZMP references over the horizon N. The complete gain sequences are computed once offline. This anticipatory structure allows the CoM to smoothly shift in advance of each footstep, producing natural and dynamically consistent gait trajectories. The resulting CoM position, velocity, and ZMP sequences are stored as discrete arrays in the microcontroller memory.

3.5 Inverse Kinematics and Joint Trajectory Generation

The offline-computed CoM trajectories are converted into the joint-space using an analytical geometric solver. The bipedal platform has 6 degrees of freedom per leg (hip roll, hip pitch, hip yaw, knee pitch, ankle pitch, ankle roll), System has 12-DOF total.



Given the planned CoM position and the according to foot positions from the footstep planner, the relative position of each foot with respect to the torso is computed. The knee joint angle θ_{knee} is obtained using the law of cosines applied to the upper and lower leg link lengths L_1 and L_2 :

$$\theta_{knee} = \pi - \cos^{-1} \left(\frac{L_1^2 + L_2^2 - d^2}{2L_1L_2} \right) \quad (10)$$

where d is the Euclidean distance from hip to ankle. The remaining hip and ankle angles are resolved using atan2-based geometric decomposition in the sagittal and frontal planes [Sun,2011]. The resulting joint angle sequences θ_k for all 12 joints are pre-computed, and stored on the microcontroller as tables indexed by gait phase.

3.6 Online Push Recovery via LQR

During gait cycle external disturbances (force impact) leads the actual CoM state to deviate from the offline trajectory. The state error vector is defined as:

$$\mathbf{e}_k = \mathbf{x}_{actual,k} - \mathbf{x}_{nominal,k} = \begin{bmatrix} \Delta x_k \\ \Delta \dot{x}_k \end{bmatrix} \quad (11)$$

An LQR controller is designed for the reduced second-order LIPM model around the nominal trajectory. The optimal gain matrix $\mathbf{K}$ is obtained by solving the DARE:

$$\mathbf{P} = \mathbf{A}^T \mathbf{P} \mathbf{A} - \mathbf{A}^T \mathbf{P} \mathbf{B} (\mathbf{B}^T \mathbf{P} \mathbf{B} + R)^{-1} \mathbf{B}^T \mathbf{P} \mathbf{A} + \mathbf{Q} \quad (12)$$

$$\mathbf{K} = (R + \mathbf{B}^T \mathbf{P} \mathbf{B})^{-1} \mathbf{B}^T \mathbf{P} \mathbf{A} \quad (13)$$

where $\mathbf{Q}$ and R are the state and input weighting matrices chosen to balance recovery speed against joint effort. The resulting weight matrices used in this study are:

$$\mathbf{Q} = \begin{bmatrix} q_1 & 0 \\ 0 & q_2 \end{bmatrix}, R = r \quad (14)$$



The online corrective input is then computed as:

$$u_{corr,k} = -Ke_k \quad (15)$$

This corrective jerk command is integrated nominal joint commands retrieved from the lookup table. Since the operation in equation (15) is a 1×2 matrix-vector multiplication, the computational cost is negligible, enabling control loop execution at 500 Hz on the microcontroller.

3.7 State Estimation via Complementary Filter

The CoM state error in equation (11) is not directly measurable. It must be estimated by reading IMU data. The MPU6050 provides raw 3 axis accelerometer (a) and gyroscope (ω) measurements at 1 kHz. A discrete time complementary filter is implemented to fuse both signals and estimate the tilt angle θ :

$$\hat{\theta}_k = \alpha(\hat{\theta}_{k-1} + \omega_k T) + (1 - \alpha)\theta_{acc,k} \quad (16)$$

where $\theta_{acc_k} = \text{atan2}(a_y, a_z)$ is the accelerometer derived tilt estimate. T is the sampling time period, and $\alpha \in (0,1)$ is the filter coefficient which determines frequency. The angular velocity estimated by gyroscope after compensation. The CoM position and velocity deviations are then approximated from the tilt angle and angular velocity through the LIPM kinematic relationship:

$$\Delta x \approx z_c \cdot \hat{\theta}, \Delta \dot{x} \approx z_c \cdot \hat{\omega} \quad (17)$$

These estimates form the state error vector fed into the LQR controller at each control cycle.



3.8 Hardware Implementation

Our architecture is deployed on a custom built bipedal platform based on the Limo chassis, actuated by 12 servo motors. All onboard computation is performed by a microcontroller (ARM Cortex-M7, 600 MHz, 1 MB RAM). The communication between the microcontroller and servos is handled via a serial PWM interface at 50 Hz. The IMU data reading and complementary filter run at 500 Hz, while the LQR correction loop operates at the same rate. On the parameter table key hardware and software parameters of the platform demonstrated in Table 1 .

Parameter Description	Symbol	Value	Unit
Total Degrees of Freedom	DoF	12	-
Nominal Center of Mass (CoM) Height	z_c	0.32	m
Thigh Link Length	$A_{\{leg\}}$	0.20	m
Shank Link Length	$B_{\{leg\}}$	0.20	m
Hip Lateral Offset	$D_{\{right\}}/D_{\{left\}}$	0.05	m
Step Length (Forward)	d_x	0.08	m
Foot Step Height	H_{step}	0.04	m
Step Duration	T_{step}	0.55	s
MPC Sampling Time Period	Δt_{mpc}	0.01	s
ZMP Preview Window Horizon	$N_{preview}$	400	steps
Applied External Disturbance (Torso Push)	F_{push}	30.0	N
Disturbance Duration	Δt_{push}	0.10	s

Table 1. Hardware & Software Parameter Table

4 Electromechanical System Architecture

4.1 Mechanical Structure

The developed platform is a 12-DoF bipedal robot consisting of two identical 6-DoF legs. Each leg includes Hip Yaw, Hip Roll, Hip Pitch, Knee Pitch, Ankle Pitch, and Ankle Roll joints. The foot geometry defines the support polygon used during single- and double-support phases and provides the physical basis for ZMP-based locomotion. The structural components are designed to provide sufficient rigidity while maintaining low overall mass, reducing actuator loading and improving dynamic performance.

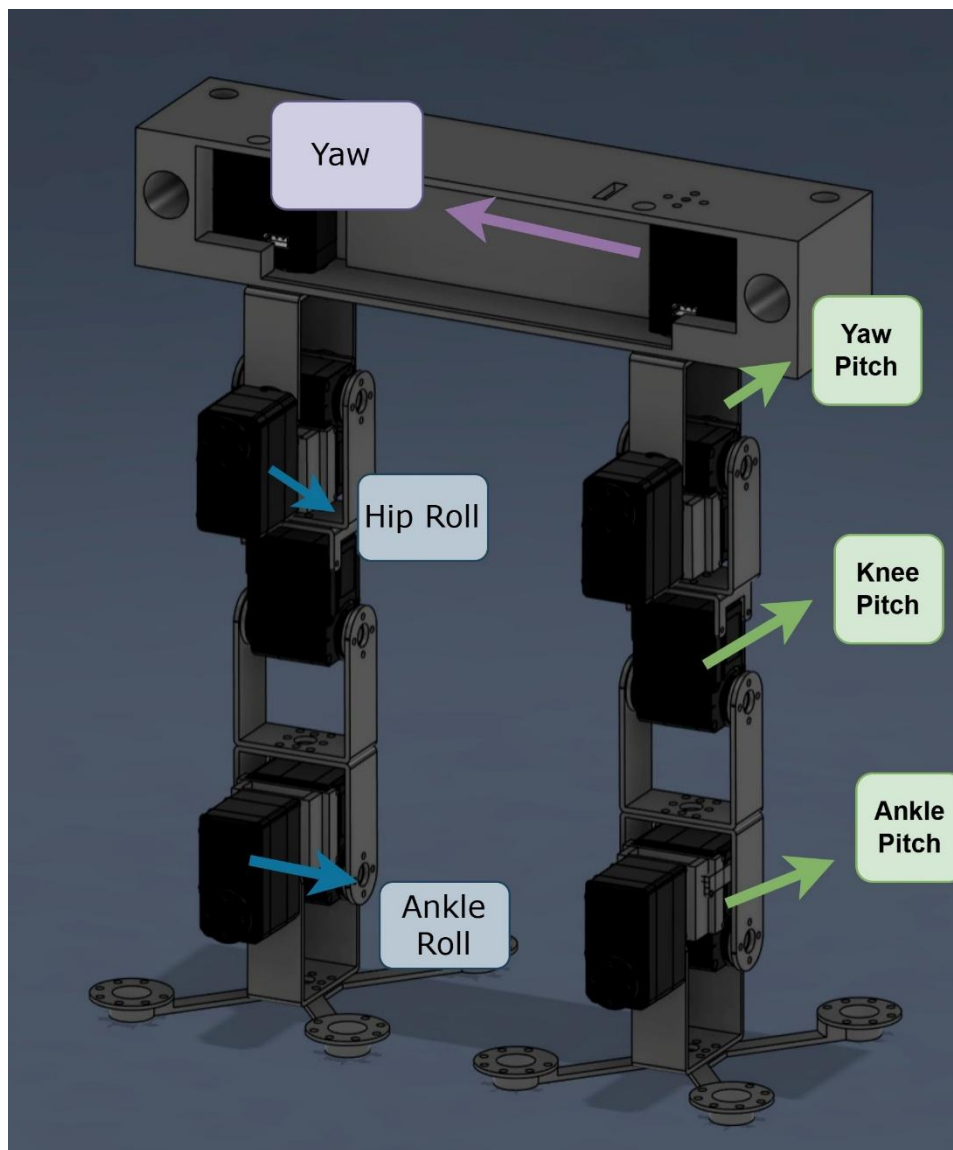


Figure 3 : The physical layout and kinematic chain hierarchy of the leg assembly, demonstrating the sequence of the Yaw, Roll, and Pitch joints from the pelvis to the foot



The proposed bipedal robot features a 12-DOF lower-body structure, with 6 DOFs allocated to each leg to ensure full spatial mobility. As illustrated in Figure 3, the kinematic chain of each leg is designed hierarchically from the pelvis (base) to the foot (end-effector) in the following sequence: Hip Yaw, Hip Roll, Hip Pitch, Knee Pitch, Ankle Pitch, and Ankle Roll.

Specifically, the Hip Yaw joint connects the torso to the leg assembly, allowing rotational movement around the vertical Z-axis. This is immediately followed by the Hip Roll (X-axis) and Hip Pitch (Y-axis) joints. The knee acts as a single-axis hinge joint driven by the Knee Pitch motor. Finally, the ankle joint combines an Ankle Pitch and an Ankle Roll motor, which is crucial for maintaining maximum foot contact with the ground and regulating the Zero-Moment Point (ZMP) during lateral disturbances.

4.2 Actuation System

All joints are actuated by serial bus servomotors controlled directly by microcontroller through a UART communication interface. Higher-torque actuators are assigned to the Knee Pitch and Hip Pitch joints, where the largest gravitational and dynamic loads occur during locomotion. The remaining joints utilize standard-torque servos sufficient for balance control and foot placement.

Each actuator provides integrated position feedback through an internal magnetic encoder, enabling CL (closed-loop) position control at the servo level. The use of serial bus communication reduces wiring complexity and allows simultaneously transmission of joint commands during gait execution.



4.3 Sensing and State Estimation Hardware

Robot state estimation is performed using an MPU6050 inertial measurement unit mounted near the torso region. The sensor provides three-axis accelerometer and gyroscope measurements that are sampled at high frequency by the microcontroller.

A complementary filter combines the short-term accuracy of gyroscope measurements with the long-term stability of accelerometer measurements to estimate body orientation. These orientation estimates are subsequently used to approximate the CoM state deviations required by the LQR push-recovery controller.

To monitor foot-ground interaction, force-sensitive resistor (FSR) sensors are embedded within both feet. The sensor measurements are used to estimate the center of pressure (CoP) and to experimentally verify the ZMP behavior predicted by the controller. This information is primarily utilized for system evaluation and stability analysis.

4.4 Embedded Control Hardware

All real-time computations are executed on a microcontroller featuring a 600 MHz ARM Cortex-M7 processor. The nominal walking trajectories generated offline by the preview controller are stored in onboard memory as lookup tables and retrieved during locomotion according to the gait phase.

The online control layer executes complementary filter, state estimation, and LQR feedback calculations. Since the push-recovery algorithm requires only matrix-vector operations using pre-computed gains, the computational demand remains well within the processing capability of the embedded platform.

4.5 Power Distribution System

The robot is powered by a rechargeable 3-cell lithium-polymer battery pack. Separate power rails are employed for the actuators and low-voltage control electronics to reduce the influence of motor-induced electrical noise on sensor measurements and microcontroller operations, as illustrated in the power distribution network layout in Figure 4.

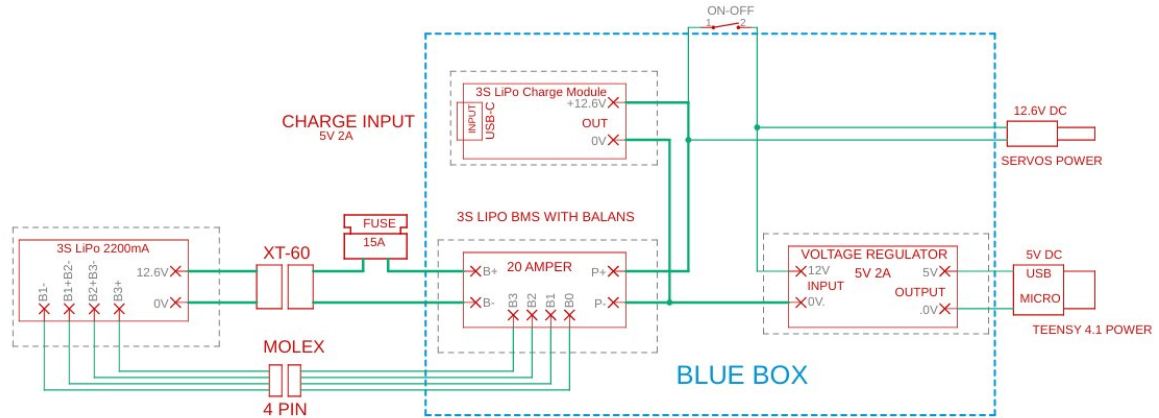


Figure 4: Power Distribution Network: 3S LiPo battery, BMS, charge module, voltage regulator, and servo power bus.

A voltage regulation stage supplies the required logic voltage for the microcontroller and sensing hardware, while the actuator bus is powered directly from the battery source [Fig 4]. Protection circuitry is incorporated to prevent over-current, over-discharge, and short-circuit conditions during operation

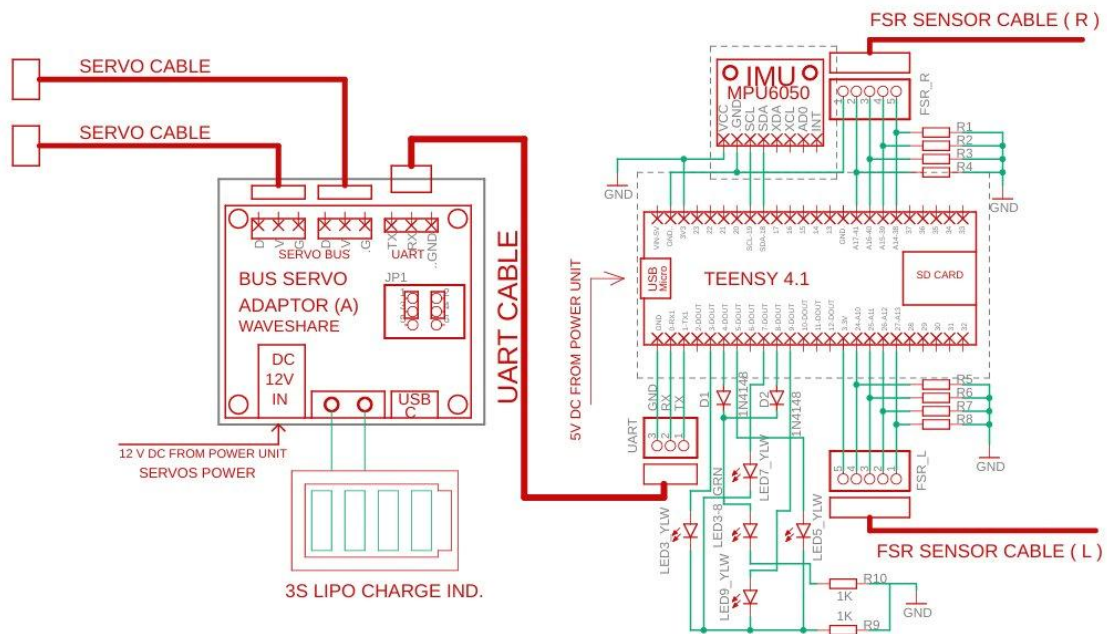


Figure 5: Control Electronics Schematic: Teensy 4.1, MPU-6050 IMU, Bus Servo Adaptor (Waveshare), FSR sensor arrays, and UART communication interface



5. SIMULATION RESULTS

The performance of the proposed low-cost control architecture was tested using a 12-DOF bipedal robot model in the MuJoCo physics engine. To test the stability of the algorithm, two different scenarios were analyzed using only Inverse Kinematics (IK)-based joint position control: 'Figure 6' nominal walking without external disturbances and 'Figure 7' push recovery using a capture point strategy under a lateral impact.

5.1. Scenario A: Nominal Walking Without External Disturbances

In the first scenario the robot was commanded to walk on a flat surface with no external forces. The nominal Center of Mass (CoM) trajectory, generated offline by the MPC algorithm, was tracked in real-time using state-space equations. However MuJoCo accurately calculate heel-strike impacts and included the mass of the robot's legs unlike the simplified LIPM assumption (which the leg masses are neglected). As a result, the simulated dynamics exhibited more realistic transient behaviors during foot-ground contact and swing-leg motion.

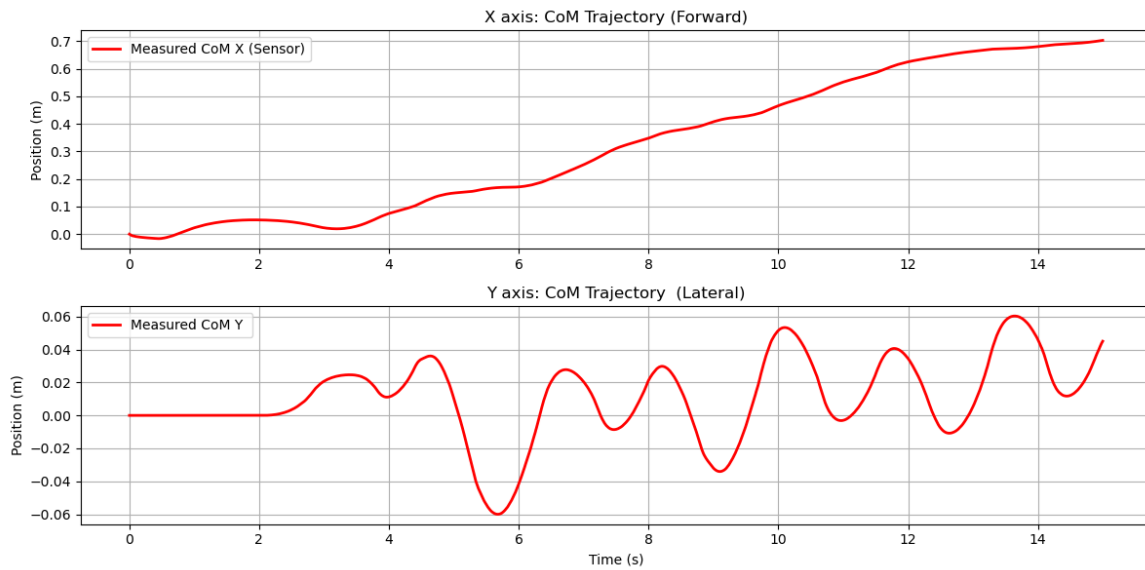


Figure 6: CoM Trajectory without Force Impact



As shown in Figure 6, the measured CoM position follows footsteps certain precision. The target joint angles computed via analytical IK were directly fed to MuJoCo's internal PD controllers. Due to the standard LIPM assumptions and the absence of an upper body to provide a compensatory inverse moment for the lower body moment, the ideal CoM trajectory deviated from the Z axis. The results demonstrate that this position-based approach is sufficient to achieve human like natural smooth walking, despite the mass distribution simplifications inherent in Linear Inverted Pendulum Model (LIPM).

5.2. Scenario B: Push Recovery via Capture Point Strategy

The second scenario tests the robustness of the system against external disturbances. During the walking cycle, a lateral impact of 30 Newtons was applied to the robot's torso for 0.1 seconds (between $t = 6.0$ and $t = 6.1$). Figure 7 illustrates the lateral CoM trajectory response during the impact. At the moment the force is applied (indicated by the vertical dotted line), the actual CoM rapidly deviates from yaw axis of the planned trajectory and accelerates. At this critical point, rather than commanding extra balancing torque to the ankle motors (ankle strategy), the system triggers a capture point reflex. The dynamic step planner instantly detects this drift through the state error and expands the target landing foot position of the swing leg toward the direction of the fall [Englsberger, 2011; Wahrmann, 2018]. As clearly seen in the graph, robot takes a wider step to prevent falling. Within a few seconds after the impact, the CoM error dissipates, and the robots returns to its nominal walking trajectory.

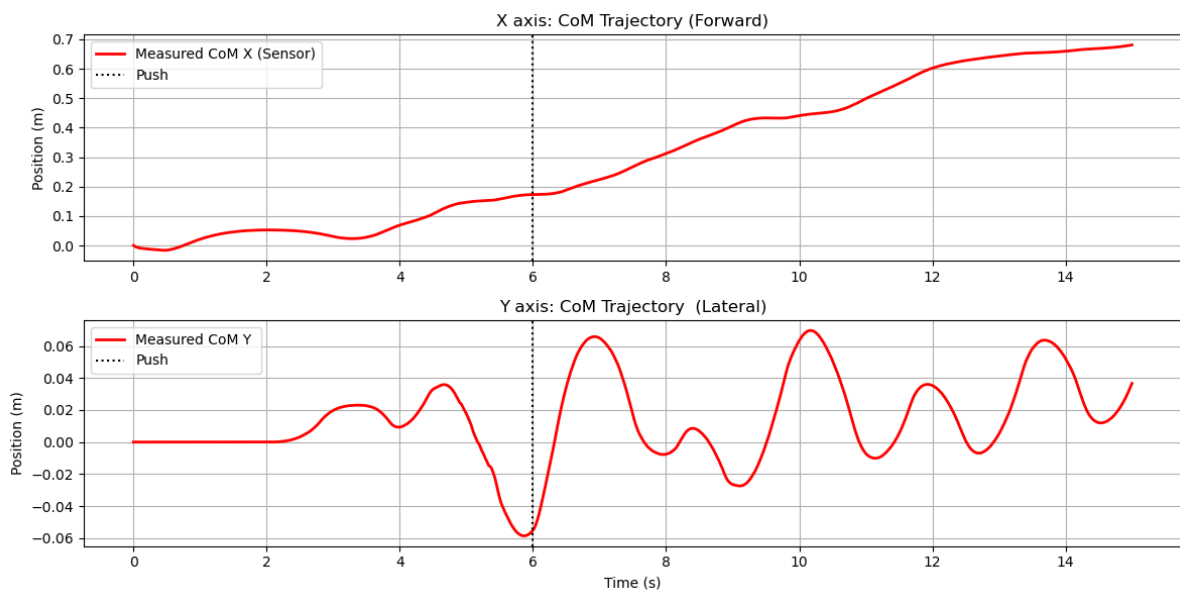


Figure 7: CoM Trajectory with Force Impact



5.3. Discussion

These findings indicate that push recovery operations in bipedal robots do not strictly require computationally expensive algorithms like Nonlinear MPC or online torque optimization by calculated by minicomputer unlike entry level microcontroller.

In the proposed architecture, heavy trajectory planning is solved offline, while instantaneous balance reactions are handled simply by updating the kinematic target points (capture point). The joint actuators only follow position references. This strategic division of tasks drastically reduces the overall computational cost of the system. Consequently, it provides a practical and industrial framework that can easily run on low-budget microcontroller, for edge computing applications.

6. CONCLUSION AND FUTURE WORK

In this study, a computationally efficient and low-cost control architecture for 12-DOF bipedal robots was proposed and successfully validated in the MuJoCo physics engine. Unlike traditional methods that heavily rely on real-time nonlinear torque optimization requiring expensive hardware, the proposed framework utilizes an offline Model Predictive Control (MPC) for nominal trajectory generation. For dynamic balance, an online capture point strategy combined with Inverse Kinematics (IK) position control was implemented to achieve robust push recovery. The simulation results demonstrated that the robot could maintain stable walking and successfully recover from a 30 N lateral impact by dynamically adjusting its swing leg's landing position without demanding extra joint torque limits.

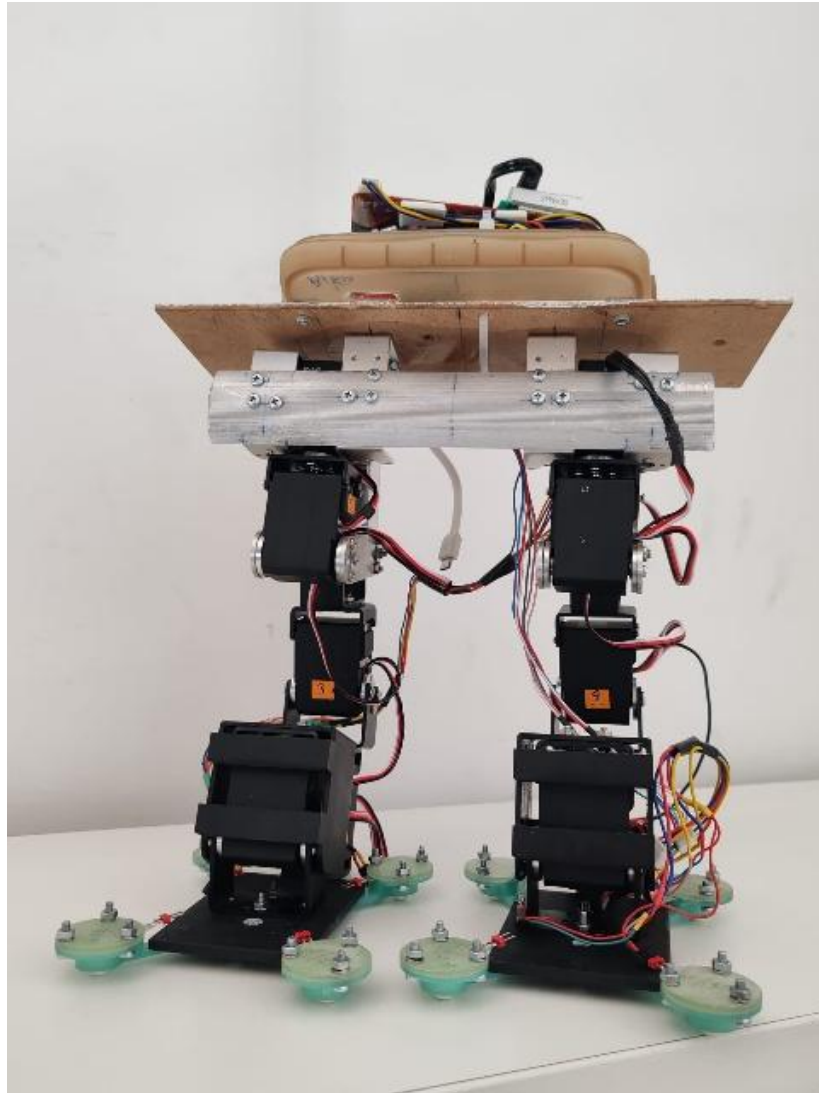


Figure 8: First Prototype of Robot (Early Phase)

Furthermore, the mechanical assembly and initial electronic wiring of the physical 12-DOF bipedal robot platform have been completed, as shown in Figure 8. The strategic division of trajectory planning and instantaneous balance reflexes drastically reduces the computational cost, making it highly suitable for edge computing. As future work, the control algorithms validated in this simulation environment will be deployed onto a low-budget microcontroller. Subsequent research will primarily focus on bridging the Sim-to-Real gap, addressing real-world hardware constraints such as sensor noise and backlash, to achieve autonomous physical locomotion.



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